

Customer Segmentation
Task 2 – Customer Segmentation Analysis
Oasis Infobyte
Submitted by: Bhavyaa Bansal
Track: Data Analytics

1. Introduction

Customer segmentation is the process of dividing customers into groups based on similar purchasing behaviour. It enables businesses to understand their customers better, develop personalized marketing campaigns, improve customer satisfaction, and increase overall profitability.

In this project, customer segmentation was performed on an e-commerce dataset using RFM (Recency, Frequency, Monetary) Analysis and the K-Means Clustering algorithm. The RFM model evaluates customers based on how recently they purchased, how frequently they purchase, and how much they spend. After creating these behavioural features, K-Means clustering was applied to identify meaningful customer groups that can support data-driven marketing strategies.

2. Objective

The primary objectives of this project were:

- To analyze customer purchasing behaviour using transaction data.
 - To calculate Recency, Frequency, and Monetary (RFM) metrics for each customer.
 - To standardize the RFM features before clustering.
 - To determine the optimal number of customer clusters using the Elbow Method.
 - To apply K-Means clustering for customer segmentation.
 - To visualize customer segments using scatter plots.
 - To profile each customer segment and recommend suitable marketing strategies.
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3. Dataset Description

The dataset contains transaction records of customers from an e-commerce platform. Each record represents a purchase made by a customer and includes demographic information, purchase details, and customer behaviour attributes.

The dataset contains the following variables:

- Customer ID
- Purchase Date
- Product Category
- Product Price
- Quantity
- Total Purchase Amount

- Payment Method
- Customer Age
- Returns
- Customer Name
- Gender
- Churn

These attributes provide sufficient information to construct customer-level behavioural features for segmentation.

4. Data Preprocessing

Before performing customer segmentation, the dataset was cleaned and prepared for analysis.

The preprocessing steps included:

- Loading the dataset using the Pandas library.
- Inspecting the data structure and column information.
- Identifying missing values and verifying data quality.
- Detecting duplicate columns and removing redundant attributes.
- Converting the Purchase Date column into datetime format for time-based analysis.
- Aggregating transaction-level data into customer-level data for RFM analysis.

These preprocessing steps ensured that the dataset was accurate and suitable for clustering.

5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand customer purchasing patterns before applying machine learning techniques.

The following descriptive statistics were calculated:

- Average Purchase Value
- Purchase Frequency
- Customer Lifetime Value (CLV)

Several visualizations were created to understand the data distribution, including:

- Purchase Frequency Distribution
- Customer Lifetime Value Distribution
- Recency Distribution
- Frequency Distribution
- Monetary Distribution
- Correlation Heatmap

The analysis indicated that most customers made relatively few purchases, while only a small proportion of customers purchased frequently and contributed a significant share of total revenue.

6. RFM Analysis

RFM Analysis is a customer segmentation technique widely used in marketing and customer relationship management.

Recency (R): Recency measures the number of days since a customer's most recent purchase. Customers with lower recency values are considered more active because they have purchased recently.

Frequency (F): Frequency represents the total number of purchases made by each customer during the available transaction period. Customers with higher frequency values demonstrate stronger engagement with the business.

Monetary (M): Monetary measures the total amount spent by each customer. Customers with higher monetary values contribute more revenue and are generally considered more valuable. These three behavioural features formed the basis for customer segmentation.

7. Feature Scaling

Since the RFM variables have different numerical ranges, feature scaling was necessary before clustering.

The StandardScaler technique from Scikit-learn was used to standardize the data. Standardization transforms each feature so that it has a mean close to zero and a standard deviation close to one.

This step prevents variables with larger numerical values, such as Monetary, from dominating the clustering process.

8. Customer Segmentation using K-Means Clustering

The K-Means clustering algorithm was applied to the standardized RFM features.

Before selecting the number of clusters, the Elbow Method was used to determine the optimal value of K. The Within-Cluster Sum of Squares (WCSS) was calculated for different values of K, and the point where the reduction in WCSS started to level off was selected as the optimal number of clusters.

After determining the appropriate number of clusters, K-Means assigned every customer to a specific segment based on similarities in purchasing behaviour.

9. Cluster Visualization

The resulting customer segments were visualized using scatter plots with different feature combinations, including:

- Frequency vs Monetary
- Recency vs Monetary

A bar chart showing the number of customers in each cluster was also generated to understand the distribution of customers across the identified segments.

Cluster profiling was performed by calculating the average Recency, Frequency, and Monetary values for every cluster.

10. Customer Segment Profiles

Based on the RFM characteristics, customers can generally be classified into the following categories:

Loyal Customers

These customers purchase frequently, spend the highest amounts, and have purchased recently. They represent the most valuable customer segment.

Potential Loyal Customers

These customers have moderate purchase frequency and spending behaviour. With personalized recommendations and targeted promotions, they have the potential to become loyal customers.

Occasional Customers

These customers purchase infrequently and spend a moderate amount. They require engagement strategies to encourage repeat purchases.

At-Risk Customers

These customers have not made purchases recently and generally exhibit low purchase frequency and lower spending. Businesses should implement retention campaigns to re-engage these customers.

11. Marketing Recommendations

Based on the customer segmentation results, the following business strategies are recommended:

Loyal Customers

- Offer loyalty rewards and membership programs.
- Provide early access to new products.
- Send personalized product recommendations.

Potential Loyal Customers

- Encourage repeat purchases using reward points.
- Recommend complementary products through cross-selling.
- Provide personalized discount offers.

Occasional Customers

- Send seasonal promotions and limited-time offers.
- Increase engagement through email marketing campaigns.
- Recommend products based on previous purchase history.

At-Risk Customers

- Launch customer win-back campaigns.
 - Offer special discount coupons.
 - Send reminder emails highlighting new arrivals and exclusive offers.
 - Collect customer feedback to understand reasons for reduced engagement.
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12. Conclusion

This project successfully demonstrated how customer segmentation can be performed using RFM Analysis and K-Means Clustering. By transforming transaction-level data into meaningful behavioural features, customers were grouped into distinct segments based on their purchasing patterns.

The analysis enables businesses to better understand customer behaviour, prioritize marketing efforts, improve customer retention, and maximize revenue through personalized strategies. Customer segmentation is an effective business intelligence technique that supports data-driven decision-making and helps organizations build stronger long-term relationships with their customers.

Future improvements may include incorporating additional behavioural features such as customer churn probability, product preferences, return behaviour, and predictive machine learning models to further enhance segmentation accuracy and business insights.

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